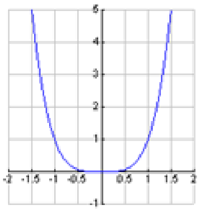
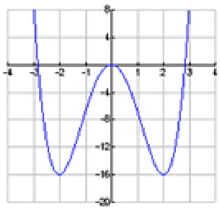
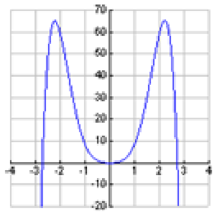
1.3: Equations and Graphs of Even and Odd Polynomial Functions

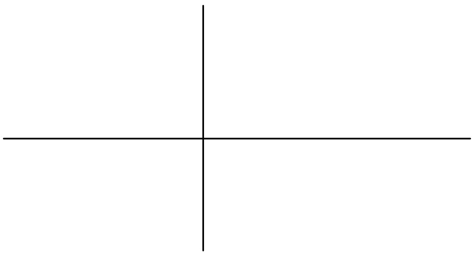
How is symmetry represented in the equation of a polynomial function? Let's investigate...

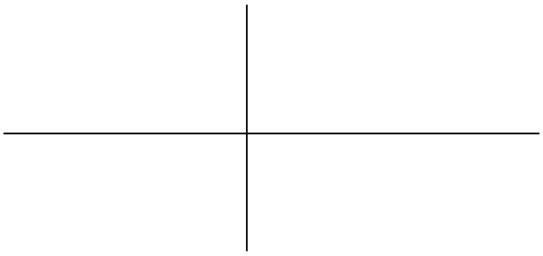
Example 1: Analyze the following even-degree polynomial functions. Determine the type of symmetry. Calculate f(-x) for each function. Compare the equations of f(x) and f(-x).

	 $f(x) = x^4$	 $f(x) = x^4 - 8x^2$	 $f(x) = -x^6 + 7x^4 + 3x^2$
symmetry:			
equation of f(-x):			
comparison:			

When a reflection in y-axis = base graph,

the function is EVEN!





Even degree

and

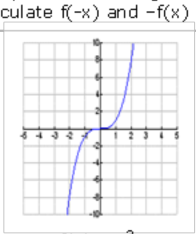
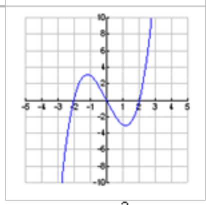
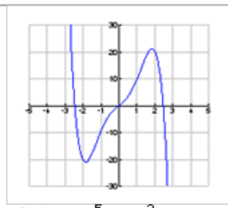
EVEN function!

Even degree

but **NOT**

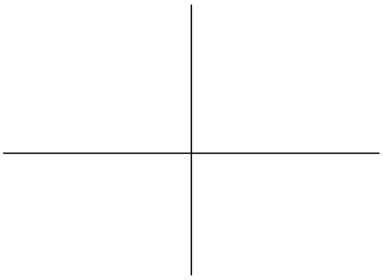
EVEN function!

Example 2: Analyze the following odd-degree polynomial functions. Determine the type of symmetry. Calculate f(−x) and −f(x) for each function. Compare the equations of f(x), f(−x) and −f(x).

	 $f(x) = x^3$	 $f(x) = x^3 - 4x$	 $f(x) = -x^5 + 5x^3 + 6x$
symmetry:			
equation of f(-x):			
equation of -f(x):			
comparison:			

When a reflection in y-axis = reflection in x-axis,

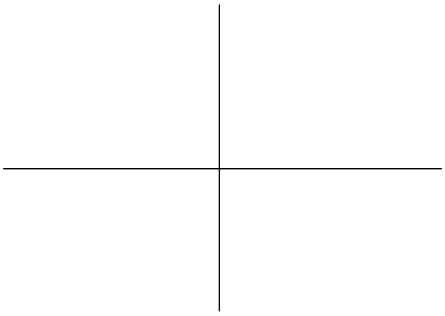
the function is ODD!



Odd degree

and

ODD function!

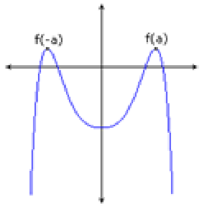


Odd degree

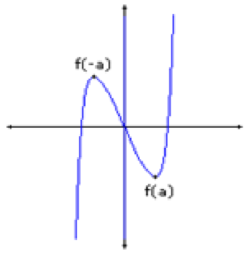
but **NOT**

ODD function!

Summary for Even and Odd Polynomial Functions:



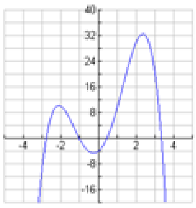
Even function:
- An even-degree polynomial function where each term of the equation has an even exponent.
- Satisfies the property f(-x) = f(x) for all x in the domain of f(x).
- Symmetric about the y-axis (line symmetry).

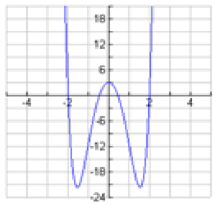


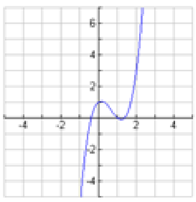
Odd function:
- An odd-degree polynomial function where each term of the equation has an odd exponent.
- Satisfies the property f(-x) = -f(x) for all x in the domain of f(x).
- Rotationally symmetric about the origin (point symmetry).

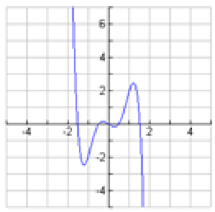
NOTE: DO NOT CONFUSE **EVEN/ODD FUNCTIONS** WITH **EVEN/ODD DEGREE FUNCTIONS!**

Example 3: Classify each function as even, odd or neither. Justify your answer.

a) 

b) 

c) 

d) 

e) $f(x) = x^5 + 6x^3 - 9x$

f) $f(x) = -x^6 + 5x^4 - x^2$

g) $f(x) = x^5 + 6x^3 + 4x^2 - 2x$

h) $f(x) = 2x(x+1)(x-2)$

i) $f(x) = x^5 + 6x^3 - 9x + 3$

j) $f(x) = -x^6 + 5x^4 - x^2 + 4$

HOMEWORK: p. 39 # 3, 4, 5, 7, 8, 12, 14ab

#3(ii) incorrect answer NOT even
#8c incorrect answer, no point symmetry